

LIFE-LINE TUNNEL RESCUE ROBOT – HOW IT MOVES (ANIMATION)

1. LOWERING THE ROBOT

The robot is lowered into the tunnel using a winch and steel cable.



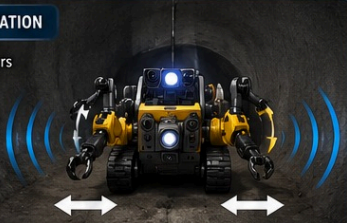
2. FORWARD MOVEMENT

The motors rotate the tracks to move the robot forward. Operator controls the speed and direction.



3. TUNNEL ADAPTATION

The ultrasonic sensors measure the tunnel width and the arms automatically adjust (widen or narrow) to fit the space.



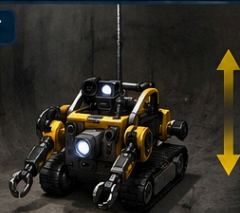
4. OBSTACLE DETECTION

Sensors detect obstacles in front and on the sides. The robot changes direction to avoid them.



5. VERTICAL MOVEMENT

The winch motor moves the robot up or down. (Useful for multi-level tunnels or shafts)



6. VICTIM DETECTION

Thermal camera and heart rate sensor detect a trapped victim.



7. AIR SUPPLY & CO₂ REMOVAL

The robot supplies O₂ through the air hose and removes CO₂ using the built-in system.



8. COMMUNICATION

Live video and audio are sent to the operator through a fiber/wired cable. No internet needed.



9. RETRIEVAL

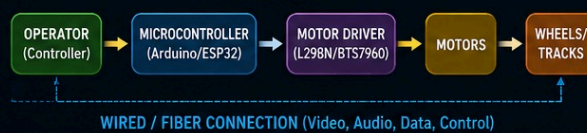
After the mission, the robot is lifted back to the surface using the winch.



MOVEMENT COMPONENTS



CONTROL FLOW



OTHER MOVEMENTS



Category	Components	Purpose
1. Sensors	Camera Module	Live video monitoring
	Microphone	Capture victim's voice and tunnel sounds
	Ultrasonic Sensors	Measure tunnel width and detect obstacles
	IR Sensors	Short-range obstacle detection
	Thermal Camera / MLX90614	Detect human body heat
	Heart Rate Sensor (MAX30102/MAX30100)	Measure victim's pulse (contact-based)
	CO ₂ Sensor (MH-Z19B)	Monitor carbon dioxide levels
	Oxygen (O ₂) Sensor	Measure oxygen concentration
	MQ Gas Sensor	Detect smoke and hazardous gases
	IMU (MPU6050/9250)	Measure robot orientation and tilt
	Load Cell	Monitor winch lifting force
2. Actuators	Limit Switches	Detect end position of arms and winch
	DC Geared Motors	Move the robot forward/backward
	Servo Motors	Move robot arms and gripper
	Winch Motor	Raise and lower the robot
	Motor Driver (BTS7960/L298N)	Control motor speed and direction
	Servo Driver (PCA9685)	Control multiple servo motors
	Air Pump	Supply oxygen through the air hose
	CO ₂ Extraction Fan/Pump	Remove carbon dioxide from the tunnel

	LED Lights	Illuminate dark tunnel environments
	Speaker	Deliver voice messages to the victim
3. Controller	ESP32 / Arduino Mega	Main controller for sensors and motors
	Raspberry Pi (Optional)	Image processing and advanced functions
4. Power Supply	Li-ion/LiPo Battery	Supply power to the robot
	Battery Management System (BMS)	Protect the battery
	Voltage Regulator	Provide stable voltages to components
	Power Switch	Turn the robot on and off
5. Body	Aluminium Chassis	Main structural frame
	Rubber Tracks	Move over rough tunnel surfaces
	Wheels/Rollers	Support tracked movement
	Steel Cable	Lift and lower the robot
	Winch Drum	Wind and unwind the lifting cable
	Protective Enclosure	Protect electronics from dust and debris
6. Communication Systems	Fiber Optic/Ethernet Cable	Transmit video, audio, sensor data, and control signals
	Camera Video Link	Send live video to the operator
	Microphone & Speaker	Two-way voice communication
	Operator Control Console (Joystick + Monitor/Laptop)	Remote control and monitoring
7. End Effectors	Robotic Arm	Reach the victim or objects

Gripper/Claw

Hold, move, or remove small obstacles

Oxygen Delivery Hose/Nozzle

Deliver oxygen directly to the trapped victim